Science Week 5: volume, pressure and the chain to the end (written)

Ember · Tuesday before the test · about 10 minutes · 7 marks Date: _____

Skill: the Because Chain. One link per mark. Every chain ends with the thing the question asks about.

[4 marks]

1. Explain how the diaphragm and ribs cause air to move into the lungs.

[3 marks]

2. A syringe is sealed at the end so no air can get in or out. The plunger is pulled out, so the air inside has a bigger volume. Explain what happens to the pressure of the air inside the syringe.

Self-check (5 min)

□ Q1: four links, ending with air moving in.

□ Q2: I said bigger volume → lower pressure, and why (particles hit the walls less often).

□ Reflection: which link do I usually leave out?

Mark scheme

- **1.** Diaphragm contracts and moves down / rib (intercostal) muscles contract and ribs move up and out **(1)**; so the volume of the chest increases **(1)**; so the pressure inside the chest decreases, lower than outside **(1)**; so air is pushed (moves) into the lungs **(1)**.
- **2.** The same number of particles in a bigger volume / particles more spread out **(1)**; so they hit the walls less often **(1)**; so the pressure decreases **(1)**.

Score: _____ / 7

Questions written for Ember in AQA style, matched to the topics in the autumn science test (breathing, pressure in gases). No AQA past-paper questions used.